

Phase-22 — Uncertainty Bands

Protocol: Lock $\rightarrow$ Verify $\rightarrow$ Style. This document is the Verify pass.

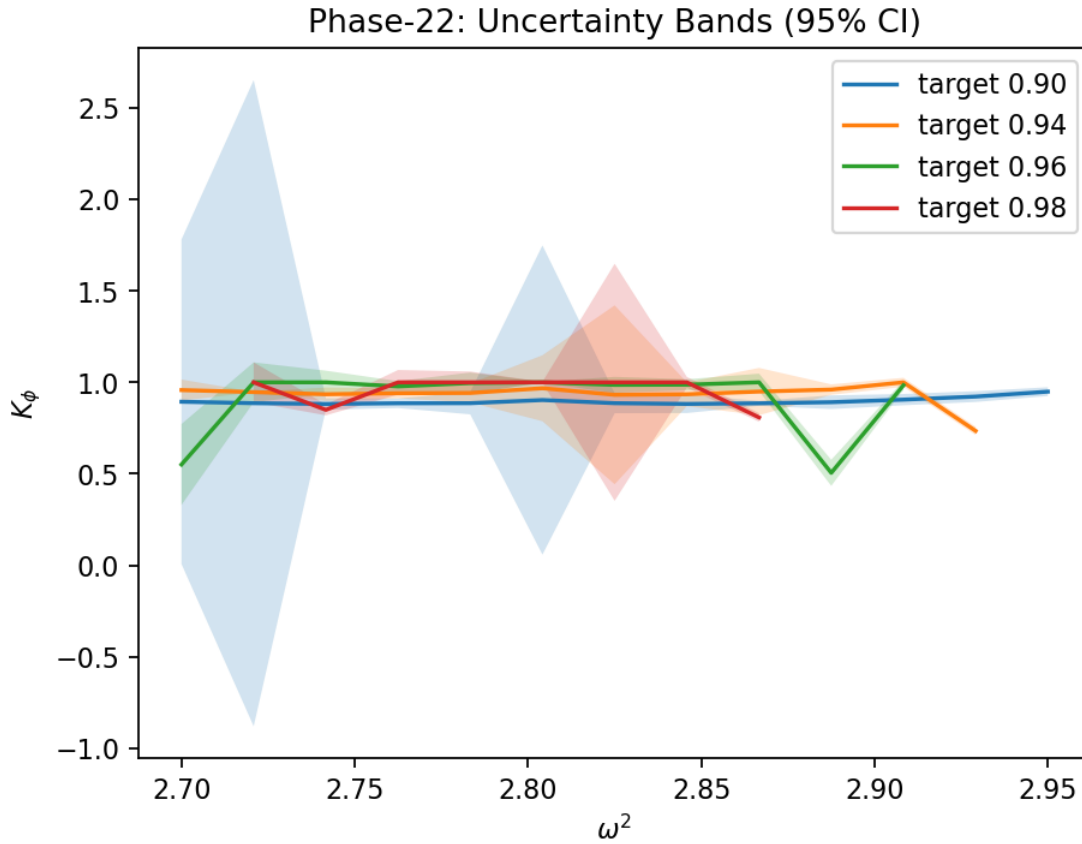
Inputs

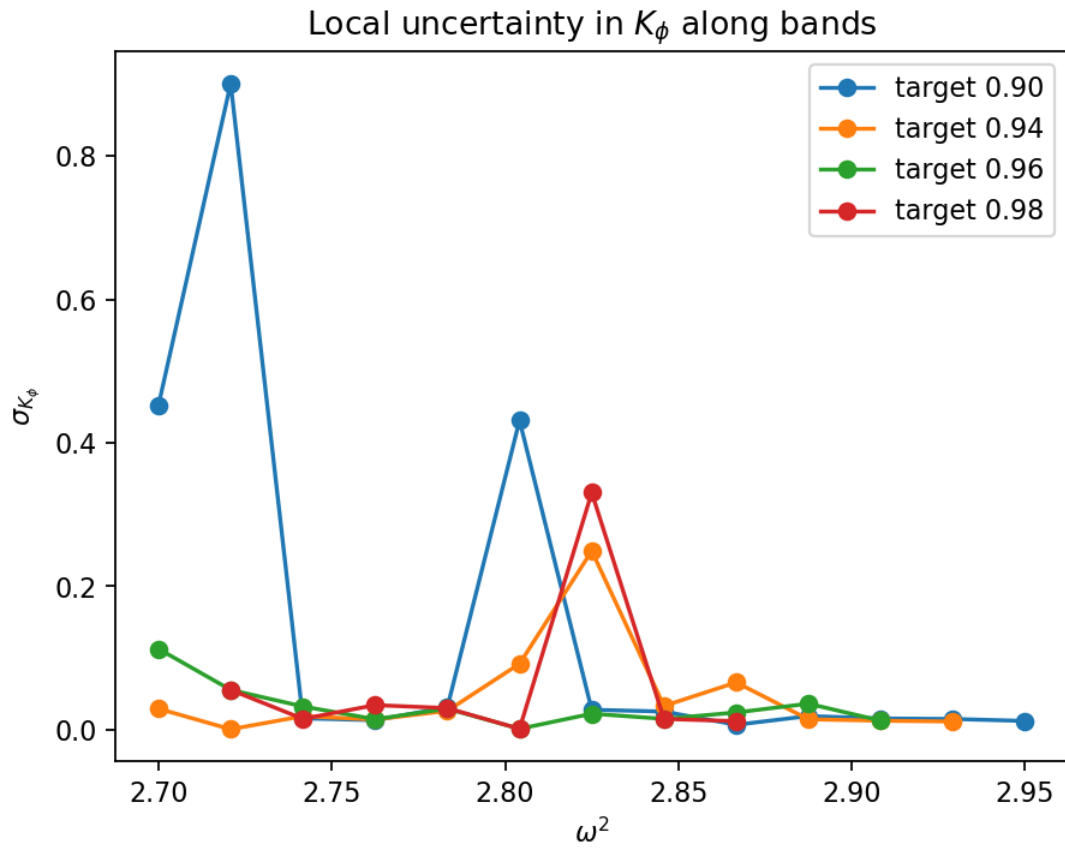
- Guardbands: from Phase-21
- Bands JSON: `outputs/phase22/p22_bands.json`

Method (summary)

- For each guardband point, estimate the local slope ($\partial s_i / \partial K_{\phi}$) via a central finite difference.
- Bootstrap the simulator at the centerline (K_{ϕ}^{mid}) to get (σ_{s_i}).
- Propagate to ($\sigma_{K_{\phi}} \approx \sigma_{s_i} / |\partial s_i / \partial K_{\phi}|$).
- Form 95% CIs for (K_{ϕ}) and band width; aggregate per target to plot ribbons.

Plots





Sample of p22_uncertainty.csv

target	omega2	Kphi_lo	Kphi_hi	Kphi_mid	si_mid	si_sd
0.9	2.7	0.18763020833333335	1.6	0.8938151041666668	0.9283342598955668	0.011186030584
0.9	2.7208333333333337	0.171875	1.6	0.8859375	0.9455858122721779	0.008453964149
0.9	2.7416666666666667	0.16328125	1.6	0.8816406250000001	0.9593759907351522	0.000532343960
0.9	2.7625	0.17097981770833331	1.6	0.8854899088541667	0.9669817052769746	0.000460526827
0.9	2.7833333333333337	0.171875	1.6	0.8859375	0.9712384694180544	0.000826930367
0.9	2.8041666666666667	0.20625000000000002	1.6	0.9031250000000001	0.9651952260481783	0.015386766695
0.9	2.825	0.171875	1.6	0.8859375	0.9705089883769598	0.000848601032
0.9	2.8458333333333337	0.16346028645833333	1.6	0.8817301432291667	0.9652516469574978	0.000856338561
0.9	2.8666666666666667	0.17044270833333333	1.6	0.8852213541666667	0.9524420801800553	0.007477345507
0.9	2.8875	0.1826171875	1.6	0.89130859375	0.9430102153768709	0.000486029999